

Agentic Graph Reasoning: Claim-Level Structural Verification for Multi-Hop Knowledge Graph Question Answering

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Abstract

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1. Introduction

Large language models answer factual questions fluently, and often enough they answer them wrongly. The errors are systematic rather than random, and they are hard to catch from the output alone, because a fabricated answer reads in the same register as a correct one; Huang et al. [1] treat this as an intrinsic consequence of how the models are built rather than an incidental defect to be patched away. The problem sharpens once an answer requires composing several facts rather than recalling one. Errors compound, since a wrong resolution at the first hop rarely announces itself downstream, and a question-level correctness check never sees the intermediate steps at all — which leaves a system that answers correctly by accident indistinguishable from one that answers correctly by sound reasoning.

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Retrieval-augmented generation [2] addresses the first half of this by grounding the model in retrieved text. It does not address the second. Retrieval runs once, before reasoning begins, so a multi-hop question must be answered from whatever that single query returned, and the passage holding the *intermediate* entity need not resemble the original question — which is the only signal the retriever has to rank on. Representing the corpus as a knowledge graph [3] makes composition a traversal rather than an inference over prose, and GraphRAG [4] exploits this, but it still retrieves statically: a neighbourhood fetched in one shot cannot know what the next reasoning step will need, and is either too small to contain the answer or so large that the relevant edges are buried.

Agentic systems close that gap by interleaving the two. Think-on-Graph [5] established the pattern with model-guided beam search over the graph, Plan-on-Graph [6] added sub-objective decomposition and reflection-driven backtracking, and Reasoning-on-Graphs [7] generates grounded relation paths as explicit plans. Retrieval becomes navigation, and the system becomes an agent operating on the graph rather than a pipeline reading from it.

One property survives this progression unchanged. Every system named above emits whatever its final generation call produces from the traversed context, and nothing checks that what the answer *asserts* is supported by what the agent actually retrieved. The gap is not fabrication — our measurements confirm that graph-navigating systems assert entities that exist in the graph and are reachable along the paths they walked (Section 5.4). It is that an entity can sit in the retrieved subgraph, have been retrieved correctly, and

still be the wrong answer: an intermediate node from the reasoning chain, a container where the question asked for a member, or one of several plausible neighbours chosen arbitrarily. A system that verifies retrieval but never verifies assertion cannot catch either case.

This paper presents **Agentic Graph Reasoning** (AGR), a knowledge-graph question-answering framework built as an explicit state machine over a constrained, deterministic graph tool API, in which the program orchestrates the loop and the language model is consulted only at bounded decision points. Its distinguishing component is a *Structural Verification Layer*: before an answer is emitted, the draft is decomposed into atomic claims and each is checked against the triples the agent actually traversed. Claims that cannot be grounded trigger targeted re-exploration or are removed, and every answer AGR returns is paired with the supporting triples that justify it.

We are precise about what that layer buys, because the ablation of Section 6 is precise about what it does not. Claim verification does not measurably change how often AGR is right; it changes how often AGR is right *for a reason it can exhibit*. The contribution is the output contract — an answer together with the traversed evidence for it — rather than an accuracy gain, and Section 6 reports the null that establishes this at the same length as the effects it does detect.

On 400 certified questions from each of WebQSP [8] and ComplexWebQuestions [9], over a Freebase-derived environment of 2.59 million entities and 8.31 million triples, AGR reaches Hits@1 of 0.755 and 0.522 against four baselines sharing one frozen backbone and one call budget, and it does so at roughly half the language-model calls of the strongest of them. On Complex-

WebQuestions it is the only system whose accuracy rises with hop count. Its advantage over the agentic baseline is located rather than asserted: on the questions where the shared cap lets that baseline finish it is slightly ahead, and the entire aggregate margin comes from the 29% and 44% of questions on which the cap truncates it.

Contributions.. This paper contributes:

1. **AGR**, a state-machine agent over a constrained graph tool API whose answers carry the traversed triples that support them, so the reasoning is auditable rather than merely asserted (Section 3).
2. A **Structural Verification Layer** that checks atomic claims against traversed adjacency before emission, with its repair routes and their measured firing rates reported as they behave rather than as designed (Section 3.2).
3. A **controlled comparison** against four baselines on one frozen backbone under one call budget, so that differences are attributable to architecture rather than to model capacity (Section 5).
4. A **component-level ablation** with paired significance testing, which finds that removing the planner *improves* accuracy on the shallower benchmark while reducing token use, and that three of four components show no detectable effect (Section 6).
5. A **hand-read census** of all 259 remaining failures, naming the *echo attractor* — a true, grounded entity one hop from the answer that every grounding check passes — and quantifying label defects in both benchmarks (Section 7).

Section 2 positions AGR against prior work. Section 3 specifies the framework and the verification layer, Section 4 the environment and experimental protocol, and Section 5 the measurements. Section 6 attributes them to components, Section 7 reads what remains, and Section 8 states the limits.

2. Related Work

3. The AGR Framework

3.1. State Machine and Tool API

3.2. Structural Verification Layer

3.3. Budgets and Termination

4. Experimental Setup

4.1. Knowledge Environment

4.2. Benchmarks, Baselines, and Backbone

4.3. Metrics and Protocol

5. Results

5.1. Accuracy

5.2. Cost and the Location of the Margin

5.3. Accuracy by Hop Depth

5.4. Groundedness

6. Component-Level Attribution

6.1. When Planning Hurts

6.2. Three Components with No Detected Effect

7. Error Analysis

7.1. The Echo Attractor

7.2. Benchmark Label Defects

8. Discussion and Limitations

9. Conclusion

Declaration of competing interest

The authors declare no competing financial interests or personal relationships that could have influenced the work reported here.

Data availability

Code, prompts, per-question run records, and the analysis scripts that produce every figure and table in this paper are available at <https://github.com/sakif-khan/agent-ic-graph-reasoning> under the MIT licence. The knowledge environment is derived from Freebase [3] and is rebuilt from its public source by the scripts in that repository rather than redistributed; the question sets are the public WebQSP [8] and ComplexWebQuestions [9] releases.

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